

One Tune, Many Stories

Tune Reuse and Topic Diversity in Early Modern English Ballads

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Abstract

This study examines how selected ballad tunes connect different topics and whether lyrics associated with the same tune share more vocabulary. A small sample from the English Broadside Ballad Archive (EBBA) contains 101 candidate records and yields 64 distinct eligible lyrics. Its tune–topic network contains 43 topic categories and 106 links. Effective topic diversity ranges from 5.03 for Packington’s Pound to 19.53 for Come Live with Me. Shared-tune pairs have a mean cosine similarity of 0.2674, compared with 0.0922 for other pairs; a document-level permutation test gives a one-sided reference $p = 0.001$. Compressing near-duplicate lyrics at a similarity threshold of 0.90 reduces the difference from 0.1753 to 0.0876. Topic breadth and clusters of closely related texts both contribute to the observed pattern of tune reuse.

1 Introduction

A song’s words and melody often seem inseparable, making me curious about how one tune can carry different stories. This question connects my interest in music with a broader curiosity about how familiar forms support new ideas.

The English Broadside Ballad Archive (EBBA) provides lyrics, catalog descriptions, and standardized tune information for exploring this question[1, 2]. I examine how widely shared tunes span different topics and how closely related versions influence their lyrics’ apparent similarity.

The study combines topic diversity, tune–topic networks, and lexical similarity across six tune categories and 64 distinct eligible lyrics. Figure 1 summarizes corpus selection. The source materials follow EBBA’s attribution and noncommercial-use policy[4].

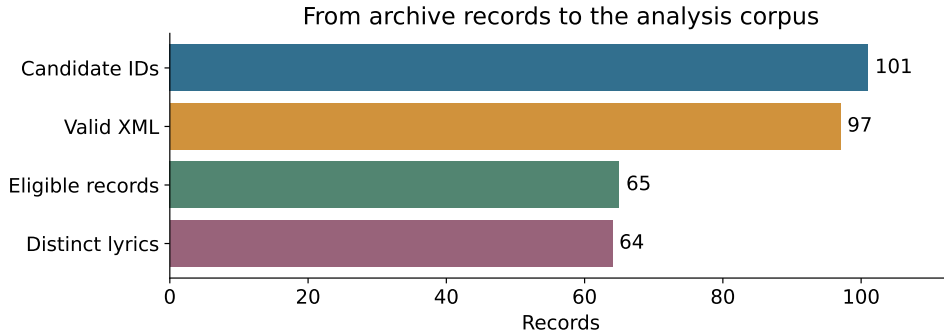


Figure 1: Corpus selection. The largest reduction follows the requirement for assigned archive topic labels.

Table 1: Corpus composition and analytical scale. Token counts refer to normalized lyrics.

Measure	Count	Measure	Count
Candidate records	101	Unavailable responses	4
Parseable XML records	97	Records without topic labels	32
Eligible records	65	Distinct eligible lyrics	64
Selected tunes	6	Observed topic categories	43
Tune–topic links	106	TF–IDF vocabulary	2,671
Median lyric length	815.5	Lyric length range	325–2,001

2 Text preparation and analytical approach

2.1 Topics, tune membership, and repeated lyrics

EBBA’s assigned keywords describe the principal cataloged subjects[3]. I extract record-level labels and standardize case, spaces, and punctuation. Two documented label pairs merge: economics/trade with economics/commerce, and religious types & sects with religious groups. Tune membership comes from the selected standardized search categories. A shared-tune pair therefore shares at least one *selected* category.

The model uses verse lines, excluding titles, tune notes, and catalog labels. Normalization preserves historical spelling and supplied letters within editorial brackets. Records with identical normalized lyrics collapse to one representative: the lowest EBBA ID supplies the topic labels, while the group retains all observed selected-tune memberships. A six-record catalog comparison confirms all tune links but only three topic sets. The analysis consistently uses the downloaded XML labels.

2.2 Topic weights and diversity

Each lyric contributes a total weight of one within each associated tune, divided equally among its assigned topics. Let T_b and K_b denote the selected tunes and topics of record b . The weight linking tune t and topic k is

$$w_{tk} = \sum_{b:t \in T_b} \frac{\mathbf{1}\{k \in K_b\}}{|K_b|}. \quad (1)$$

A two-topic lyric contributes 0.5 to each topic, keeping its total weight equal to one. With $p_{tk} = w_{tk} / \sum_j w_{tj}$, topic entropy and effective topic diversity are

$$H_t = - \sum_{k:p_{tk}>0} p_{tk} \log p_{tk}, \quad D_t = \exp(H_t). \quad (2)$$

Three equally weighted topics give $D_t = 3$; concentration lowers this value. Multiple labels per lyric allow D_t to exceed the lyric count.

Five hundred within-tune bootstrap resamples provide percentile ranges for D_t . A second comparison draws eight records without replacement from each tune with at least five eligible records, repeated five hundred times. These procedures describe variation within the observed corpus.

2.3 Lexical similarity and comparison experiments

TF-IDF represents each lyric using individual words, English stop-word removal, a minimum document frequency of two, logarithmic term frequency, and L_2 normalization[5]. Cosine similarity s_{ij} measures proximity between the 2,671-term lyric vectors. The comparison statistic is

$$\Delta = \frac{1}{|P_{\text{same}}|} \sum_{(i,j) \in P_{\text{same}}} s_{ij} - \frac{1}{|P_{\text{other}}|} \sum_{(i,j) \in P_{\text{other}}} s_{ij}, \quad (3)$$

where the two pair sets share, or do not share, a selected tune. All pairs receive equal weight. The reference distribution randomly reassigns complete tune-membership sets to documents 999 times, preserving group sizes and keeping text vectors fixed. Its one-sided reference value is $(1 + \#\{\Delta^* \geq \Delta\})/1000$. Document-level reassignment retains dependence from shared lyrics.

For near-duplicate sensitivity, a graph joins lyrics whose similarity reaches 0.90 or 0.95. Each connected component retains its lowest-ID record and that record’s metadata. The comparison then reuses the original vectors. Every random procedure uses seed 42.

3 Topic diversity across tunes

Figure 2 and Table 2 show contrasting topic distributions. Come Live with Me has 25 labels and effective diversity of 19.53, followed by Chevy Chase (18.18) and Greensleeves (16.06). Packington’s Pound has nine labels and diversity of 5.03, indicating stronger concentration. Children in the Wood’s broad label set describes only two lyrics.

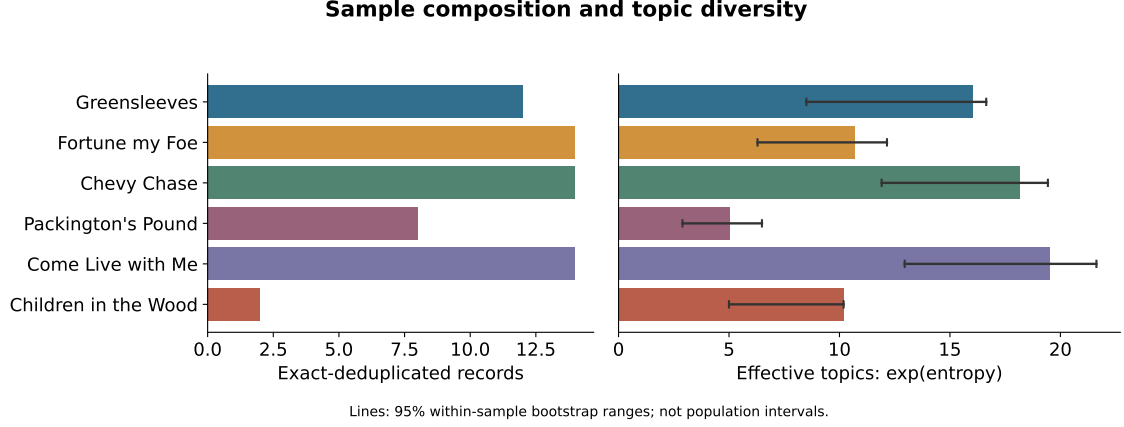


Figure 2: Record counts and effective topic diversity. Lines show within-sample bootstrap percentile ranges.

Table 2: Topic diversity after exact deduplication; ranges summarize 500 bootstrap resamples.

Tune	Records	Labels	D_t	2.5%–97.5% range
Greensleeves	12	21	16.06	[8.50, 16.64]
Fortune my Foe	14	17	10.71	[6.29, 12.15]
Chevy Chase	14	23	18.18	[11.91, 19.43]
Packington’s Pound	8	9	5.03	[2.89, 6.49]
Come Live with Me	14	25	19.53	[12.95, 21.63]
Children in the Wood	2	11	10.19	[5.00, 10.19]

At eight lyrics per tune (Figure 3; Table 3), Come Live with Me remains more diverse than Packington’s Pound. Overlapping ranges discourage a complete ranking. Packington’s Pound’s eight records yield identical subsamples; Children in the Wood is excluded.

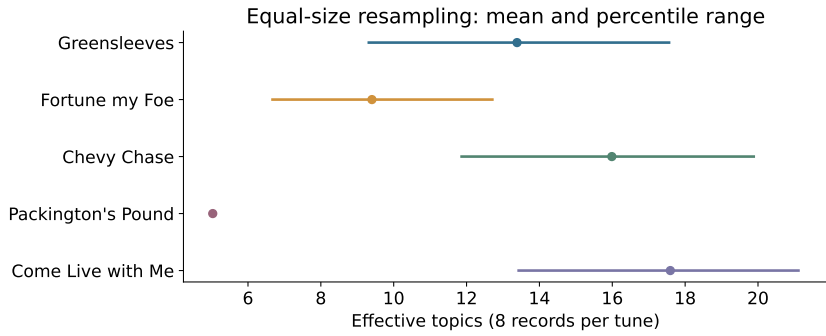


Figure 3: Equal-size resampling. Points show mean diversity; lines show percentile ranges.

Table 3: Five hundred subsamples of eight lyrics per tune, drawn without replacement.

Tune	Sample size	Mean D_t	2.5%–97.5% range
Greensleeves	8	13.39	[9.32, 17.56]
Fortune my Foe	8	9.40	[6.67, 12.71]
Chevy Chase	8	15.99	[11.86, 19.88]
Packington’s Pound	8	5.03	[5.03, 5.03]
Come Live with Me	8	17.59	[13.43, 21.11]

4 Shared topics and contrasting profiles

The network contains 106 links between six tunes and 43 observed topic categories. Figure 4 displays the fourteen topics with the greatest total weight. The crossings show how several tunes connect to common subjects while each tune combines a different set of topics. All 43 categories contribute to the diversity estimates.

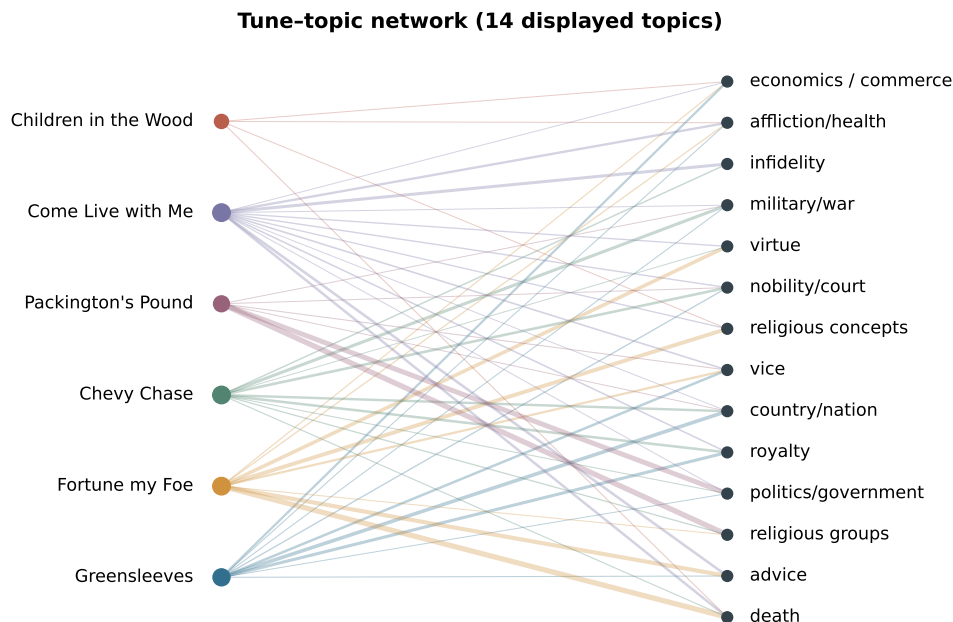


Figure 4: Tune–topic network. Edge width represents fractional record weight; shared connections indicate overlapping cataloged subjects.

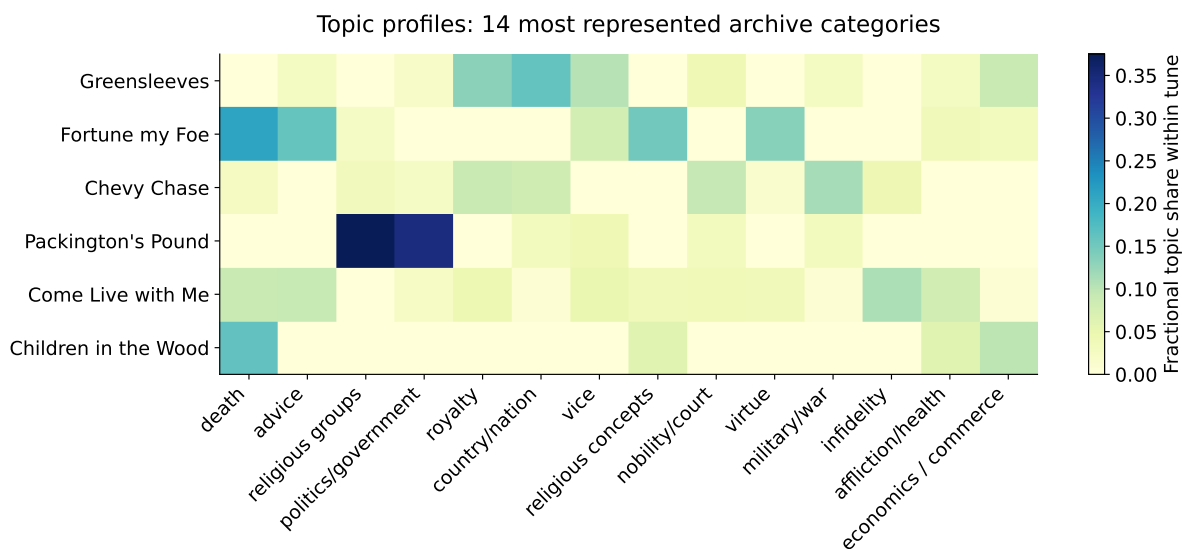


Figure 5: Within-tune shares for the same fourteen topics. Omitted topics account for the remainder of each row.

Figure 5 shows the strength of each association. The religious groups and politics/government categories together contribute 5.75 of Packington’s Pound’s total weight of eight, or 71.9%. This concentration explains its low effective diversity. The broader distribution for Come Live with Me corresponds to its higher diversity. The network describes which associations occur, while the heatmap shows how much each association contributes within a tune.

5 Lyric similarity and the permutation comparison

Shared-tune pairs have a mean cosine similarity of 0.2674, compared with 0.0922 for the other pairs. Figure 6 shows overlap at low similarity and a group of highly similar shared-tune pairs near the upper end of the scale. The difference in means is therefore compatible with both broadly shared vocabulary and a smaller cluster of closely related texts.

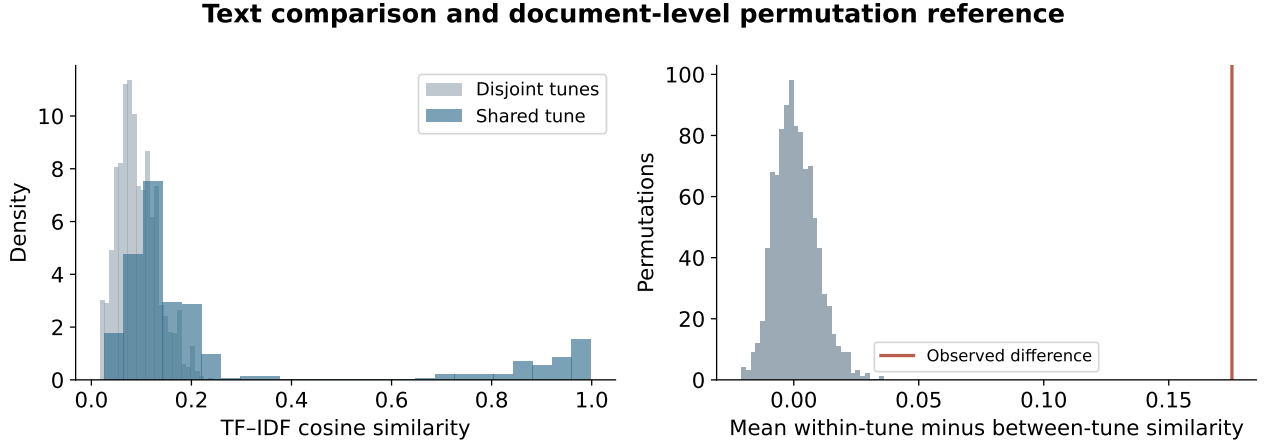


Figure 6: Left: pairwise cosine similarity distributions. Right: the document-level permutation distribution and the observed difference in means.

Table 4: Pairwise comparison and document-level permutation statistics.

Statistic	Value
Shared-tune pairs	368
Other pairs	1,648
Mean similarity: shared tune	0.2674
Mean similarity: other pairs	0.0922
Observed difference, Δ	0.1753
Document-level permutations	999
Mean permuted difference	0.0007
Permutation 2.5%–97.5% range	$[-0.0140, 0.0191]$
One-sided reference p	0.001

Table 4 gives the numerical comparison. The permutation distribution centers close to zero, and its central 95% lies between -0.0140 and 0.0191 . The observed difference of 0.1753 exceeds all 999 permuted values. The plus-one calculation gives $p = 0.001$, the smallest value this experiment can report.

This comparison asks whether the observed alignment between tune membership and vocabulary exceeds the alignment obtained by random reassignment within this corpus. It supports a sample-level association between tune categories and textual similarity. Larger groups contribute more pairs because the statistic weights pairs equally. The density plot also shows why a single mean requires further examination: highly similar lyrics can have a substantial influence on the difference.

Near-duplicate sensitivity examines this concentration by retaining one representative from each cluster of closely related lyrics. The comparison keeps the lexical representation fixed and changes which records contribute to the similarity difference.

6 Vocabulary and closely related versions

Table 5 reports the twelve highest-weight words for each tune, ranked by mean TF-IDF across its lyrics. Fortune my Foe emphasizes death, repentance, and religious vocabulary; Chevy Chase includes names and hunting-related words. Packington’s Pound includes plot and popish, while Come Live with Me emphasizes love and interpersonal terms. Historical spellings and general words such as did remain in the lists, so these terms guide interpretation without serving as topic classifications.

Table 5: Highest-weight words within each tune, in descending mean TF-IDF order.

Tune	Twelve highest-weight words
Greensleeves	fine, england, i’le, alwayes, did, sit, mault, shall, new, privy, stomachs, smith
Fortune my Foe	doth, death, time, pray, repent, live, love, souls, god, shall, world, sin
Chevy Chase	did, king, piercy, dowglas, sir, chase, thy, earle, knight, thee, thou, hunting
Packington’s Pound	plot, popish, devil, scribling, fools, plots, th, false, age, design, hall, prison
Come Live with Me	love, wanton, wife, did, beauty, gentle, friends, heart, thou, time, fair, thy
Children in the Wood	wretch, dyd, pretty, deeds, babes, woman, money, infants, lend, thirty, wrath, debt

Figure 7 and Table 6 show that near-duplicate compression reduces the shared-tune similarity advantage. The 0.95 threshold retains 47 records and gives $\Delta = 0.1205$. The 0.90 threshold retains 42 records and gives $\Delta = 0.0876$, approximately half the main estimate. Both differences remain positive; these sensitivity subsets have no separate permutation tests.

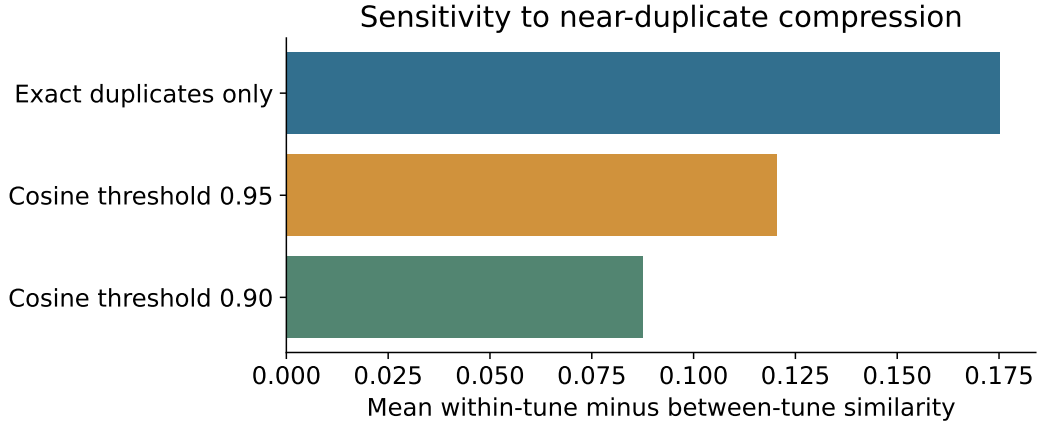


Figure 7: Sensitivity of the mean similarity difference to near-duplicate compression.

Table 6: Retained records and similarity differences under alternative duplicate treatments.

Treatment	Records	Δ
Exact duplicates only	64	0.1753
Cosine threshold 0.95	47	0.1205
Cosine threshold 0.90	42	0.0876

Packington’s Pound falls from eight records to three at both thresholds. This reduction helps explain why its lexical clustering deserves attention alongside its concentrated topic profile. Connected components identify similar wording; they do not establish a historically verified edition relationship.

7 Interpretation and conclusion

The analyses reveal two complementary features of tune reuse in this corpus. The topic network connects the same tune to different cataloged subjects, while the lyric comparisons reveal clusters of related wording within tune categories. Topic breadth and textual repetition can coexist, and each changes the interpretation of the other.

Come Live with Me illustrates topic breadth. Its effective diversity reaches 19.53 in the main corpus and averages 17.59 under equal-size sampling. Individual XML records provide concrete examples: EBBA 20655 includes advice, religious concepts, vice, and virtue, whereas EBBA 21372 includes death, infidelity, and love. These different combinations contribute to the broader profile visible in the network and heatmap. The labels describe cataloged subjects and can assign several topics to one lyric.

Packington's Pound illustrates the connection between topic concentration and textual clustering. Religious groups and politics/government account for 71.9% of its topic weight, and its high-weight words include plot and popish. Its reduction from eight records to three under both similarity thresholds indicates a concentration of closely related wording. Across the full corpus, compression at 0.90 halves the similarity difference, so the main association depends substantially on how related texts enter the sample.

I therefore interpret the results as evidence about the selected archive records: tune categories organize both diverse subjects and recurring language. Equal-size resampling preserves the broad contrast between Come Live with Me and Packington's Pound, while the duplicate comparison shows that vocabulary-based associations need to be read together with version structure. Convenience sampling and incomplete topic metadata keep these conclusions tied to the observed corpus.

The study's central finding is that a tune's range of subjects and the distinctness of its associated lyrics measure different aspects of reuse. Combining topic diversity, network structure, lexical summaries, and duplicate sensitivity gives a more informative account of how one tune can accompany multiple stories.

References

- [1] Patricia Fumerton, director. *English Broadside Ballad Archive*. University of California, Santa Barbara. <https://ebba.english.ucsb.edu/>.
- [2] English Broadside Ballad Archive. *TEI-XML*. <https://ebba.english.ucsb.edu/page/tei-xml>.
- [3] English Broadside Ballad Archive. *Keywords*. <https://ebba.english.ucsb.edu/page/keywords>.
- [4] English Broadside Ballad Archive. *Use Policy*. <https://ebba.english.ucsb.edu/page/use-policy>.
- [5] scikit-learn developers. *TfidfVectorizer*. https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html. Version used: 1.8.0.